

Eigenvalues & the characteristic polynomial

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A - \lambda \cdot I_2 = \begin{bmatrix} a-\lambda & b \\ c & d-\lambda \end{bmatrix}$$

$$\parallel$$
$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix}$$

$$T: \mathbb{F}^2 \longrightarrow \mathbb{F}^2$$

$$e_1 \mapsto \begin{bmatrix} a \\ c \end{bmatrix}$$

$$e_2 \mapsto \begin{bmatrix} b \\ d \end{bmatrix}$$

$$\begin{bmatrix} y_1 \\ y_2 \end{bmatrix} \mapsto \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

$$= \begin{bmatrix} ay_1 + by_2 \\ cy_1 + dy_2 \end{bmatrix}$$

Observation: TFAE:

$$(1) \quad \exists v \neq 0 \in F^2 \quad \& \quad \lambda \in F$$
$$\text{s.t.} \quad T(v) = \lambda v$$

$$(2) \quad \exists v \neq 0 \in F^2 \quad \& \quad \lambda \in F$$
$$\text{s.t.} \quad (T - \underbrace{\lambda \cdot \text{id}_{F^2}})(v) = 0$$

$$(3) \quad \ker(T - \lambda \cdot \text{id}_{F^2}) \neq \{0\}$$

$$(4) \quad \text{RREF}(A - \lambda \cdot I_2) \text{ has } c \text{ free variable}$$

$$(5) \quad \det(A - \lambda \cdot I_2) = 0$$

$$(6) \quad (a - \lambda)(d - \lambda) - bc = 0$$

$$\lambda^2 - \underbrace{(a+d)}_{\text{tr}(A)} \lambda + \underbrace{(ad-bc)}_{\det(A)}$$

$\text{Tr}(A)$ $\det(A)$
Defⁿ ① $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ The trace of
A is the scalar

$$\text{Tr}(A) = a + d \in F$$

② $x^2 - \text{Tr}(A)x + \det(A)$
is the characteristic polynomial
or char poly of A

③ The eigenvalues of A
are the zeros of its
characteristic polynomial

Observation repackaged:

$Av = \lambda v$ has a non-zero solution

$\Leftrightarrow \lambda$ is an eigenvalue for A.

Example

$$(1) \quad A = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\text{Tr}(A) = 2$$

$$\det(A) = 1$$

$$\begin{aligned} \text{Char poly}(A) &= x^2 - 2x + 1 \\ &= (x-1)^2 \end{aligned}$$

1 is the only eigenvalue.

and we have two linearly independent solutions to $I_2 v = 1 \cdot v$

$$(2) \quad A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \quad \text{with } Ae_1 = e_1$$

$$\text{Tr}(A) = 2, \quad \det(A) = 1$$

$$\text{Char poly}(A) = (x-1)^2$$

$$Av = 1 \cdot v \quad v = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\Leftrightarrow \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} a \\ b \end{bmatrix}$$

$$\Leftrightarrow (A - 1 \cdot I) \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Leftrightarrow \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

This only has a 1-dim'l space of solutions
 $= \left\{ \begin{bmatrix} a \\ 0 \end{bmatrix} : a \in F \right\}$

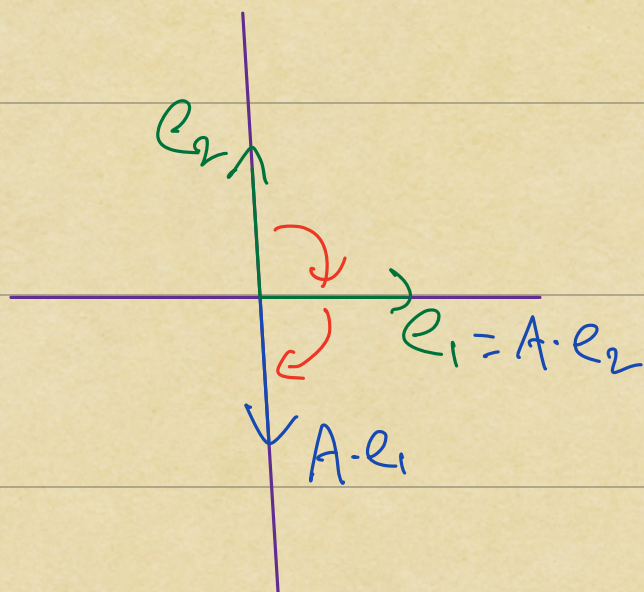
The linear transformation is not diagonalizable.

$$(3) \quad A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \in M_{2 \times 2}(\mathbb{R})$$

$$\text{Tr}(A) = 0, \quad \det(A) = 1$$

$$\text{char poly}(A) = x^2 + 1$$

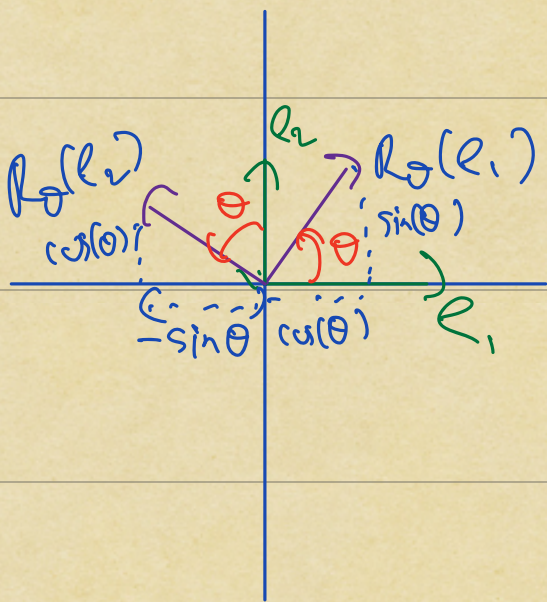
This has no zeros in $\mathbb{R}$ & so
A has **no eigenvalues** in $\mathbb{R}$



Geometrically,
A corresponds
to clockwise rotation
by $\pi/2$

(4) For any angle θ we can look at counterclockwise rotation by θ

$$R_\theta: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$



Matrix for R_θ

$$\begin{bmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{bmatrix}$$

$$\cos^2 \theta + \sin^2 \theta$$

Char poly: $x^2 - 2\cos(\theta)x + 1$

Zeros:
$$\frac{2\cos(\theta) \pm \sqrt{4\cos^2(\theta) - 4}}{2}$$
$$= \cos(\theta) \pm \sqrt{-\sin^2(\theta)}$$

$$= \cos(\theta) \pm i \sin(\theta)$$

$$= e^{\pm i\theta}$$

NOT REAL UNLESS $e^{i\theta} = \pm 1$

$$(5) \quad A = \begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

$$v_1 = \begin{bmatrix} \sqrt{3} - 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} -(\sqrt{3} + 1) \\ 2 \end{bmatrix}$$

$$T(v_1) = (2 + \sqrt{3})v_1$$

$$T(v_2) = (2 - \sqrt{3})v_2$$

$$\Rightarrow [T]_{\mathcal{B}} = \begin{bmatrix} 2 + \sqrt{3} & 0 \\ 0 & 2 - \sqrt{3} \end{bmatrix}$$

$$[T]_{\mathcal{B}}^3 = \begin{bmatrix} (2 + \sqrt{3})^3 & 0 \\ 0 & (2 - \sqrt{3})^3 \end{bmatrix}$$

$$\rightarrow [T]_{\mathcal{B}}^{10} = \begin{bmatrix} (2 + \sqrt{3})^{10} & 0 \\ 0 & (2 - \sqrt{3})^{10} \end{bmatrix}$$

A^{10} ?

$$T^{10}(v_1) = (2 + \sqrt{3})^{10} v_1$$

$$T^{10}(v_2) = (2 - \sqrt{3})^{10} v_2$$

To compute A^{10} , we need to know what T^{10} does to the standard basis.

$$A^{10} = \underbrace{C}_{\uparrow} [T]_{\mathcal{B}}^{10} C^{-1}$$

"Change of "
basis matrix
independent of powers